Bandit LAB - short script (read aloud)

10 slides, about 5 min 20 s of speech + the demo. Each speaker clicks through their own slides.

Speaker 1 - slides 1-3

Slide 1

Good afternoon, everyone. We're Bandit LAB. Have you ever stood at a busy crossing, trying to decide when it's safe to go? That's exactly the problem in our paper, “Social Attention for Autonomous Decision-Making in Dense Traffic”, from NeurIPS 2019. It teaches a self-driving car to make that decision.

Slide 2

Here's the situation. Our car wants to turn left, and there's no traffic light. Cars are coming from every side. If it goes too early, it crashes. If it waits too long, it just sits there forever. We can't write a rule for every possible case, so instead we let the car learn from experience, by trial and error. That's reinforcement learning.

Slide 3

So how does it learn? Every second, the car looks around: where it is, how fast it's going, and the same for up to fourteen cars nearby. Then it makes a simple choice: slow down, keep its speed, or speed up. If it drives smoothly and quickly, it gets a reward. If it crashes, it loses five points. Over time, the car learns a score, called Q, for each choice. This formula simply says: a good choice now gives a good reward now and leads to a good situation next. Now I'll hand over to my teammate, who'll explain how the car actually learns this score.

Speaker 2 - slides 4-6

Slide 4

Thanks. Now, there are far too many situations to write all the scores in a table. So we use a neural network to estimate them. This is called DQN, a Deep Q-Network. To keep the learning stable, it uses three simple tricks. First, it remembers past moments and studies them in random order, a bit like revising old notes. Second, it checks its answers against a slowly updated copy of itself, so the target doesn't keep moving. And third, at the start it tries random moves, just to explore.

Slide 5

Now here's the real question the paper asks: how should the car look at the other cars? It compares three ideas. FCN reads all the cars as one long list of numbers, but the list gets shuffled, so it can't tell which car is which. CNN looks at a small, blurry picture of the road from above. And attention, the paper's idea, looks at every car and asks: how important is this one for me? To keep it fair, all three networks are about the same size.

Slide 6

Attention is actually very close to how we drive. Think about it: when you cross a road, you don't look at every car equally. You focus on the one coming fast towards you. Our car does the same. It gives each car a danger score, and softmax turns those scores into percentages. Here, the fast car from the left gets seventy-two percent of the attention, and the rest get very little. And the nice part is, we can actually see what it's looking at, like the lines in this picture. Now my teammate will show you how well it works.

Speaker 3 - slides 7-10

Slide 7

Thanks. So, did it work? We trained each network three times and tested them on a hundred situations they had never seen. And just like in the

paper, attention came out on top: sixty-seven percent success, and only twelve percent crashes. FCN was too reckless; it crashed forty-one percent of the time. And CNN was too careful; it often just waited until the time ran out.

Slide 8

But we didn't want to stop there, so we pushed the idea further than the paper. First, we trained one single agent to handle all twelve routes: left, straight and right, from every road. Attention cut the crashes in half compared with FCN. Then we built a small city, six junctions joined together, with a finish line at the end. Even without any extra training, attention finished fifty-two percent of the journeys, while FCN managed only thirty-four.

Slide 9

Let's see it in action. The green band is the route it has to follow. The coloured lines show which cars it's paying attention to right now. And on the right, you can see what it decides to do, second by second.

Slide 10

So, to wrap up. The big lesson is this: how a car looks at the traffic around it matters just as much as how it learns. Attention makes it safer, it works with any number of cars, and we can actually understand its decisions. Thank you very much. We're happy to take any questions.

If the TA asks

What is DQN? It's Q-learning, but instead of a table, a neural network guesses the score of each action.

What is attention? The car gives each other car an importance score, and softmax turns those scores into percentages that add up to one.

Why doesn't the order of the cars matter? Attention just adds up all the cars with their weights, and a sum is the same in any order.

Why only three runs per network? Each run takes about an hour on a laptop. The paper used 120, so we show the uncertainty on our graphs.

Why can't it drive around obstacles? It only controls its speed, not the steering. Changing lanes would be a nice next step.